MEASURE THEORY:
AN INVITATION TO ADVANCED REAL ANALYSIS

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Abstract

These are some notes I made while reading through Folland's book [Fol99]. Real Analysis is a very difficult subject, especially at the graduate level. As such, I figured I'd make them public so other learners have more resources to study.

Hopefully, graduate students and those studying for PhD qualifying examinations in Analysis will find them useful.

Comments and suggestions to improve this manuscript are most welcome:

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Please also email me if you discover any errors.

Acknowledgements

I am deeply grateful to my friend Valdemar Andersen, with whom I hiked 230 kilometers from Portugal to Spain on the Camino de Porto.

"The trick with education is to teach people in such a way that they don't realise that they are learning until it's too late."- Harold Eugene Edgerton

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Date: August 2026.

1. Sigma Algebras

Here is an example of a $\sigma-$algebra:

Theorem 1. Let X be an uncountable set. Then the collection

$$\mathcal{M} = \{E \subset X : E \text{ is countable or } E^c \text{ is countable}\}$$

is a $\sigma-$algebra on X.

Proof: $\varnothing \in X$ and in particular, $\varnothing$ is countable, so that $\varnothing \in \mathcal{M}$.

Let $E \in \mathcal{M}$ be given. Now, if E is countable, $(E^c)^c = E$ is countable, so that $E^c \in \mathcal{M}$. If E^c is countable, that immediately satisfies our requirement.

To verify the last condition for a $\sigma-$algebra, let $\{E_n\}_{n=1}^{\infty}$ be a family of elements in $\mathcal{M}$. We proceed by cases:

Case 1: All the E_n's are countable

Since the countable union of countable sets is countable[1], then

$$\bigcup_{n=1}^{\infty} E_n \in \mathcal{M}$$

by definition.

Case 2: At least one E_n is only co-countable.

Then, since

$$E_n \subset \bigcup_{k=1}^{\infty} E_k,$$

taking complements yields

$$\left(\bigcup_{k=1}^{\infty} E_k\right)^c \subset E_n^c$$

so by assumption on E_n, $(\bigcup_{k=1}^{\infty} E_k)^c$ is a subset of a countable set, and is therefore countable. That is,

$$\bigcup_{k=1}^{\infty} E_k$$

is co-countable (and therefore in $\mathcal{M}$).

Cases 1 and 2 highlight that $\bigcup_{k=1}^{\infty} E_k$ is always contained in $\mathcal{M}$, so that by definition, $\mathcal{M}$ is a $\sigma-$algebra on X. ■

[1]See Theorem 1.5.8 (ii) in [Abb15]

2. Measures

One of the biggest problems in Mathematics is finding a way to quantify size. Abstract *distances* can be understood in terms of metrics, which were then used to develop the theory of functions on metric spaces; see Chapter 19 of [Tao16] for a quick review.

We now turn to the issue of finding an abstract notion of *size*, or rather, *volume*.

Just as we defined a *metric* $d : X \times X \to [0, \infty]$ on a set X to satisfy certain properties[2] that are in accordance with the familiar notion of distance in Euclidean space, we wish to define a generalised "*volume*" function $\mu : \mathcal{M} \to [0, \infty]$[3] over a $\sigma-$algebra $\mathcal{M}$ which assigns an abstract volume $\mu(E)$ to a set $E \in \mathcal{M}$; recall that the $\sigma-$algebra is a family of sets.

Now, how do we generalise the size of a set? That is to say, given a σ-algebra $\mathcal{M}$, what properties should a function $\mu : \mathcal{M} \to [0, \infty]$ have?

For anything to be endowed with the responsibility of giving a generalised volume to a set, it must surely assign the value "0" to the empty set $\varnothing$, as it has no elements. Intuitively, if you imagine an empty water tank, then the amount of water inside, according to any reasonable scale of measurement, must be zero.

We therefore impose our first condition on μ: $\mu(\varnothing) = 0$.

What are some other desirable properties? Consider a sequence $\{E_k\}_{k=1}^{\infty}$ of disjoint sets in the $\sigma-$algebra. We ought to have the relation

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} E_k.$$

To see why this must hold intuitively, consider cutting a cake into however many parts you like. Then certainly the sum of "μ" of all the slices would equal "μ" of the entire cake, which is equivalent to the *union* of all the pieces!

[2]Symmetry, non-degeneracy and the Triangle Inequality.

[3]We relegate the image of μ to the nonnegative real line as it doesn't make sense for an object to possess negative volume or if you wish, size.

We might wish to impose more requirements on our generalised volume function μ, but as demonstrated on page 20 of [Fol99] and 2.22 in [Axl20], some definitions will be inconsistent with one another, and in fact, the two above already give a rich theory. We therefore require μ to satisfy the two conditions discussed above, and call such functions a *measure*:

Definition: Measures

Let X a set equipped with a $\sigma-$algebra $\mathcal{M}$. A *measure* on $\mathcal{M}^a$ is a function $\mu : \mathcal{M} \to [0, \infty]$ such that

(1) $\mu(\varnothing) = 0$
(2) If $\{E_k\}_{k=1}^{\infty}$ is a sequence of disjoint sets in $\mathcal{M}$, then $\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k)$.

[a] If $\mathcal{M}$ is clear from the context, we may simply say "on X."

(2) in the definition above immediately implies

$$\mu\left(\bigcup_{k=1}^{n} E_k\right) = \sum_{k=1}^{n} \mu(E_k),$$

for a finite collection of disjoint sets $E_1, \dots, E_n \in \mathcal{M}$, because we may take $\{F_k\}_{k=1}^{\infty}$ to be

$$F_k = \begin{cases} E_k & , 0 \leq k \leq n, \\ \varnothing & , \text{otherwise.} \end{cases}$$

For X a set and $\mathcal{M} \subset \mathscr{P}(X)$ a $\sigma-$algebra, we call the pair $(X, \mathcal{M})$ a *measurable space*. Sets contained in $\mathcal{M}$ are referred to as *measurable*. If μ is a measure on $(X, \mathcal{M})$, then the triple $(X, \mathcal{M}, \mu)$ is called a *measure space.*

Here is an example of a measure, as outlined in this theorem.

Theorem 1. Let X be an uncountable set, and let $\mathcal{M}$ be the $\sigma-$algebra of countable or co-countable sets.[4]

Define the function $\mu : \mathcal{M} \to [0, \infty]$ by

$$\mu(E) = \begin{cases} 0, & E \text{ countable} \\ 1, & E \text{ co-countable,} \end{cases}$$

where $E \subset X$. Then μ constitutes a measure on $(X, \mathcal{M})$.

Proof: The empty set is countable, so $\mu(\varnothing) = 0$. To prove countable additivity, consider first a disjoint sequence $\{E_k\}_{k=1}^{\infty}$ of countable sets in X. The countable union of countable sets is countable, so that

[4] This was discussed in Section 1, Theorem 1.

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = 0 = \sum_{k=1}^{\infty} \mu(E_k).$$

Now, if we had some $j \in \mathbb{N}$ such that E_j is co-countable, the union $\bigcup_{k=1}^{\infty} E_k$ is also co-countable (see Case 2 in Section 1, Theorem 1). Moreover, disjointness implies that $E_k \subset E_j^c$ for all $k \neq j$, i.e., the E_k's are subsets of a countable set for $k \neq j$, and are therefore countable. By definition of μ,

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = 1, \qquad \sum_{k=1}^{\infty} \mu(E_k) = \mu(E_j) = 1,$$

so in general, we have shown, for a disjoint sequence $\{E_k\}_{k=1}^{\infty}$,

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k),$$

that is, μ possesses countable additivity. ■

Measures have some useful properties, as outlined in this theorem.

Theorem 2 (Folland 1.8).
Let $(X, \mathcal{M}, \mu)$ be a measure space. Then,

(1) (*Monotonicity*)
If $E, F \in \mathcal{M}$ and $E \subset F$, then $\mu(E) \leq \mu(F)$.

(2) (*Subadditivity*)
If $\{E_k\}_{k=1}^{\infty} \subset \mathcal{M}$, then $\mu\left(\bigcup_{k=1}^{\infty} E_k\right) \leq \sum_{k=1}^{\infty} \mu(E_k)$.

(3) (*Continuity from below*)
If $\{E_k\}_{k=1}^{\infty} \subset \mathcal{M}$ and $E_1 \subset E_2 \subset \dots$, then $\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \lim_{k\to\infty} \mu(E_j)$.

(4) (*Continuity from above*)
If $\{E_k\}_{k=1}^{\infty} \subset \mathcal{M}$ and $E_1 \supset E_2 \supset \dots$, and $\mu(E_1) < \infty$, then $\mu\left(\bigcap_{k=1}^{\infty} E_k\right) = \lim_{k\to\infty} \mu(E_k)$.

The proof of this result can be found on page 27 of [Fol99].

The measure of two sets is analogous to the dimension of the sum of two finite-dimensional vector subspaces; first recall the familiar theorem from linear algebra[5]

Dimension of Two Subspaces.
Let V_1 and V_2 be subspaces of a finite-dimensional vector space. Then

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2).$$

[5] For a proof, see 2.37 in [Axl24]

If we view the dimension as the size of the vector space, then the formula above is reminiscent of how (long ago) we were taught to find the area of two intersecting shapes: You first find the sum of their two individual areas, and then subtract the area of the region where they overlap.

A similar result holds for measures:

Theorem 3.
If $(X, \mathcal{M}, \mu)$ is a measure space and $E, F \in \mathcal{M}$, then

$$\mu(E \cup F) = \mu(E) + \mu(F) - \mu(A \cap F),$$

provided $\mu(E \cap F) < \infty$.

To prove this result, we need a lemma:

Lemma 1. Suppose $(X, \mathcal{M}, \mu)$ is a measure space and $E, F \in \mathcal{M}$ are such that $E \subset F$. Then, if $\mu(E) < \infty$, $\mu(F \setminus E) = \mu(F) - \mu(E)$.

Proof: As $F = E \cup (F \setminus E)$ and this union is disjoint, $\mu(F) = \mu(E) + \mu(F \setminus E)$, so since $\mu(E) < \infty$, we may subtract $\mu(E)$ from both sides to obtain

$$\mu(F \setminus E) = \mu(F) - \mu(E),$$

as desired. $\square$

Proof of Theorem 3: We may write

$$E \cup F = (E \setminus (E \cap F)) \cup (F \setminus (E \cap F)) \cap (E \cap F).$$

Since the union above is disjoint, finite additivity of the measure implies

$$\mu(E \cup F) = \mu(E \setminus (E \cap F)) + \mu(F \setminus (E \cap F)) + \mu(E \cap F).$$

The intersection $E \cap F$ is a subset of *both* E and F, so invoking Lemma 1,

$$\begin{aligned} \mu(E \cup F) &= (\mu(E) - \mu(E \cap F)) + (\mu(F) - \mu(E \cap F)) + \mu(E \cap F) \\ &= \mu(E) + \mu(F) - \mu(E \cap F) \end{aligned}$$

which is the desired result. $\blacksquare$

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